

MAROON CAYS PILOT CENTER PITCH

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CORE RESEARCH FOCUS

Solar Microgrid Resilience

- Performance in salt air + humid climates
- Hurricane durability test
- Battery Optimization in isolated grids

Circular waste systems

- Waste diversion and landfill reduction
- Runoff & marine pollution prevention
- Closed-loop nutrient recovery

Energy systems

- Monitored solar arrays
- Smart inverters + battery storage
- Storm impact & degradation monitoring

Waste & Water systems

- Stream-separated waste systems
- Diversion rate measurement
- Runoff & environmental impact tracking



REVENUE & PARTNERSHIP

Research Grants

- Renewable energy innovation grants
- Environmental foundations
- Sustainability research programs

University & Academic programs

- Field research residencies
- Engineering + environmental cohorts
- Lab access & hosting fees

Corporate & industry Testing

- Solar technology pilots
- Battery storage testing
- Waste system innovation trials

Demonstration & training hub

- Infrastructure workshops
- Climate adaptation training
- Policy + planning site visits





CURRENT SOLAR LANDSCAPE IN THE BAHAMAS

Energy Structure

- Heavy diesel dependence, imported fuel volatility, high electricity prices (2–3x U.S.)

Recent Solar Expansion

- 20–25 MW utility-scale PPAs
- 25 MWp Coral Harbour project (~53 GWh/year)
- 11 MW Grand Bahama project

Gap

- Generation-focused, no island-based resilience testing, limited pilot validation infrastructure

SOLAR RESEARCH FOCUS & DIFFERENTIATION

Research Focus

- Hurricane-rated mounting systems, high-UV exposure durability, battery storage + microgrid integration

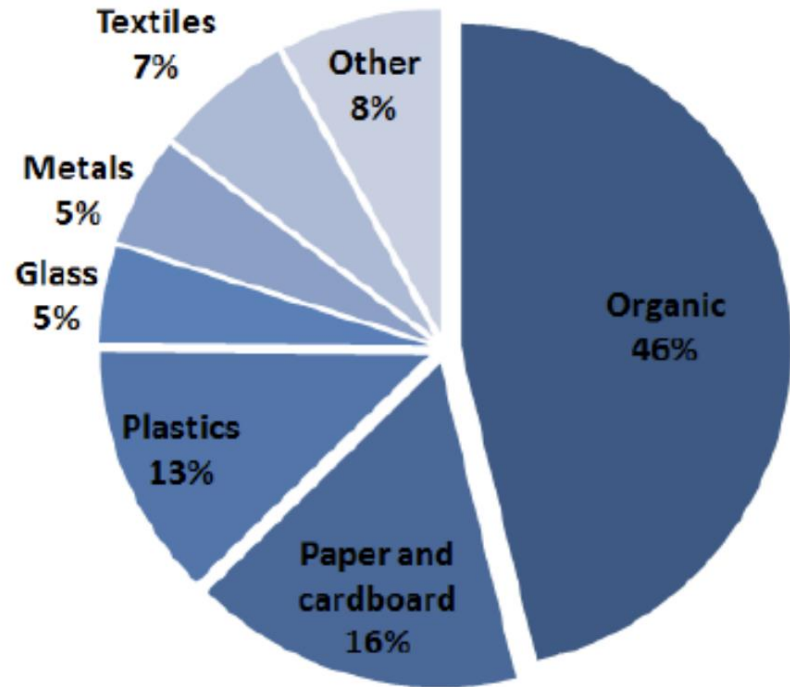
Differentiation

- Real hurricane-zone deployment, live island microgrid testbed, climate resilience + commercialization focus, phased pilot deployment model

Island-based solar resilience lab — first of its kind in the region



WASTE MANAGEMENT: CHALLENGE OR OPPORTUNITY?



1. Issue: 687 million metric tons of plastic debris accumulating on its shorelines by 2025

- Threat: Displacement of biomass of the people within these islands.

2. Issue: Dumping trash illegally.

- Threat: Dumped waste leaches hazardous substances contaminating the water and air.

3. Issue: 138,000 tons of organic waste in tourism-related solid waste

- Threat: health hazard

Figure 1: Average composition of waste in Bahamas (source UNEP, 2017)

WASTE TO ENERGY RESEARCH CENTER

Core research questions and starting points

- What proportion of municipal solid waste in Exuma is suitable for energy conversion (specifically organic waste)
- What are the environmental and economic returns of implementing an anaerobic digestion + biodiesel hybrid model?





RISKS & MITIGATIONS

- **1. Regulatory & Infrastructure Delays**
 - Early government alignment
 - Local logistics partner: Exuma Waste Management
 - Energy consultation: Stellar Energy Bahamas
- **2. Harsh Climate Technology Degradation**
 - Corrosion-resistant systems (Sunfolding)
 - Containerized microgrids (BoxPower)
- **3. Capital Intensity / Funding Risk**
 - Federal grant applications
 - Public–private partnership strategy

PARTNERSHIPS



PARTNER WITH
UCHICAGO

UChicago – Environmental Frontiers

- *Set special topic projects on the island*

PARTNERSHIPS WITH OTHER UNIVERSITIES

Stanford – Natural Capital Alliance

University of Bahamas

Project Overview



Innovations in Climate Resilient Coastal Zones - Overview (The Caribbean) >



TECHNOLOGY PARTNERSHIPS

Solar Panel Startups

- BoxPower, EssInc



Waste Startups

- Caribshare, Streamline IntegratedEnergy



MAROON CAY PITCH



WHO WE ARE

- Island-based solar + waste research & pilot hub
- Real-world tropical stress-testing environment
- Microgrid demonstration site
- Long-duration storage validation (iron flow)
- Coastal-resilient solar innovation testing
- Modular waste-to-energy integration
- University + startup + industry collaboration model



An aerial photograph of a tropical coastline, featuring a sandy beach, turquoise water, and lush greenery. A large white circle is superimposed on the left side of the image, containing the text 'MARKET OPPORTUNITY' in bold, black, sans-serif capital letters.

MARKET OPPORTUNITY



TARGET PARTNERSHIPS

1. Non-profit

a. be Waste Wise (bWW)

- i. Brings the best minds together to build a global waste management community.

2. Start ups

a. New Providence Ecology Park

- i. Focused on constructing an E-waste facility.

3. Community facilities

a. The Exuma Foundation

- i. Hosting people when the foundation runs programs

CASE STUDY: COST AND BENEFIT OF WASTE TO ENERGY DIGESTOR

Data and economic results analysis

Description	Unit	Values
Generator cost estimation	USD	674,171.00
Biodigester cost estimation	USD	23,809.52
Maintenance cost	USD	269.67
Electricity production	kWh/y	6,570,000.00
Unitary energy cost	USD/kWh	0.10
Biomass used	kg/month	4,500,000
Investment	USD	713,000.00
Financial rate	%, a.y.	7.15
Instalments	USD/y	102,220.20
Net present value (NPV)	USD	3,087,820.53
Return on investment (ROI)	%	433%
Payback	Year	1.01

This is a table detailing the costs and benefit trajectory of using a biodigester from the case study in São Paulo City

GROWTH POTENTIAL

Advanced Marine Conservation Research

- Mangrove storm-buffer analytics
- AI-powered reef health monitoring system
- Coral restoration & reef resilience modeling

Next Generation Energy Innovation

- Offshore micro-wind feasibility testing
- Tidal & wave energy pilot studies
- Advanced battery chemistry trials

Scalable Commercial Impact

- Regional replication model for Caribbean islands
- Infrastructure licensing opportunities
- Climate resilience consulting & training programs

Costal resilience & Infrastructure Innovation

- Climate-resilient coastal engineering models for storm-prone island systems
- Mangrove and reef-based natural defense analytics for surge and erosion protection
- Distributed renewable energy frameworks for remote coastal communities



SUMMARY

- Island-based solar + waste pilot test center
- Real tropical deployment — hurricane, salt, humidity
- Integrated microgrid and waste-to-energy systems
- Clear pathway from testing to commercialization
- Strategic access to island and coastal markets



Q & A



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